

Database Architecture

1. Overview

The database of **BuyBlink** is designed to store and manage user information, product details, orders, and transactions efficiently. A relational database model is used to ensure data consistency, integrity, and easy retrieval. The design supports core e-commerce functionalities such as user authentication, product browsing, order management, and payment tracking.

2. List of Tables

The following tables are used in the BuyBlink database:

Table Name	Description
Users	Stores user and admin details
Categories	Stores product category information
Products	Stores eco-friendly product details
Orders	Stores order information
Order_Items	Stores products associated with each order
Payments	Stores payment transaction details

3. Table Structure and Attributes

3.1 Users Table

Stores information about registered users and administrators.

Attribute Name	Data Type	Description
user_id (PK)	INT	Unique user identifier
name	VARCHAR	User full name
email	VARCHAR	User email address
password	VARCHAR	Encrypted password
role	VARCHAR	User role (User/Admin)
created_at	DATETIME	Account creation date

3.2 Categories Table

Stores product categories.

Attribute Name	Data Type	Description
category_id (PK)	INT	Unique category identifier
category_name	VARCHAR	Name of the category

3.3 Products Table

Stores details of eco-friendly products.

Attribute Name	Data Type	Description
product_id (PK)	INT	Unique product identifier
name	VARCHAR	Product name
description	TEXT	Product description
price	DECIMAL	Product price
stock	INT	Available quantity
eco_rating	INT	Sustainability rating
category_id (FK)	INT	References Categories table

3.4 Orders Table

Stores order details made by users.

Attribute Name	Data Type	Description
order_id (PK)	INT	Unique order identifier
user_id (FK)	INT	References Users table
total_amount	DECIMAL	Total order value
order_status	VARCHAR	Order status
order_date	DATETIME	Date of order

3.5 Order_Items Table

Stores items included in each order.

Attribute Name	Data Type	Description
order_item_id (PK)	INT	Unique order item ID
order_id (FK)	INT	References Orders table
product_id (FK)	INT	References Products table
quantity	INT	Quantity ordered
price	DECIMAL	Price per item

3.6 Payments Table

Stores payment transaction details.

Attribute Name	Data Type	Description
payment_id (PK)	INT	Unique payment ID
order_id (FK)	INT	References Orders table
payment_method	VARCHAR	Mode of payment
payment_status	VARCHAR	Payment status
payment_date	DATETIME	Date of payment

4. Relationships Between Tables

- A **User** can place multiple **Orders** (1-to-Many)
- An **Order** can contain multiple **Order_Items** (1-to-Many)
- A **Product** can appear in multiple **Order_Items** (1-to-Many)
- A **Category** can have multiple **Products** (1-to-Many)
- Each **Order** is associated with one **Payment**

5. ER Diagram

The ER diagram visually represents the database entities, their attributes, and relationships, ensuring clarity in data flow and structure.

